

JKBOSE 2026

CLASS 11TH

**TRIGONOMETRIC
FUNCTIONS**

MATHS

LECTURE-08

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TOPICS TO BE COVERED

1

Miscellaneous Exercise

2

Related Problems

MISCELLANEOUS EXAMPLE

Find the value of $\tan \frac{\pi}{8}$.

(Sol) we know that $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$.

Put $x = \frac{\pi}{8}$, we get

$$\tan 2\left(\frac{\pi}{8}\right) = \frac{2 \tan\left(\frac{\pi}{8}\right)}{1 - \tan^2\left(\frac{\pi}{8}\right)} = \frac{2t}{1 - t^2}, \text{ where}$$

$$t = \tan\left(\frac{\pi}{8}\right)$$

$$\Rightarrow \tan\left(\frac{\pi}{4}\right) = \frac{2t}{1 - t^2}$$

$$\Rightarrow 1 = \frac{2t}{1 - t^2}$$

$$\Rightarrow 1 - t^2 = 2t$$

$$\Rightarrow t^2 + 2t - 1 = 0$$

$$\Rightarrow t = \frac{-2 \pm \sqrt{4 - 4 \times 1 \times -1}}{2 \times 1}$$

$$\Rightarrow t = \frac{-2 \pm \sqrt{4 + 4}}{2}$$

$$\Rightarrow t = \frac{-2 \pm 2\sqrt{2}}{2}$$

$$\Rightarrow t = \frac{-1 \pm \sqrt{2}}{1}$$

$$\Rightarrow t = -1 \pm \sqrt{2}$$

$$\Rightarrow \tan\left(\frac{\pi}{8}\right) = -1 + \sqrt{2}, -1 - \sqrt{2}$$

Since $\frac{\pi}{8}$ lies in the 1st Quad.

$$\text{So, } \tan\left(\frac{\pi}{8}\right) = \sqrt{2} - 1$$

MISCELLANEOUS EXERCISE

CHAPTER – 3

QUESTION

$$* 2 \cos x \cos y = \cos(x+y) + \cos(x-y)$$

$$* \cos(\pi - \theta) = -\cos \theta$$

Prove that:

$$2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0$$

Proof: LHS = $2 \cos\left(\frac{\pi}{13}\right) \cos\left(\frac{9\pi}{13}\right) + \cos\left(\frac{3\pi}{13}\right) + \cos\left(\frac{5\pi}{13}\right)$

$$\Rightarrow \text{LHS} = \cos\left(\frac{\pi}{13} + \frac{9\pi}{13}\right) + \cos\left(\frac{\pi}{13} - \frac{9\pi}{13}\right) + \cos\left(\frac{3\pi}{13}\right) + \cos\left(\frac{5\pi}{13}\right)$$

$$\Rightarrow \text{LHS} = \cos\left(\frac{10\pi}{13}\right) + \cos\left(-\frac{8\pi}{13}\right) + \cos\left(\frac{3\pi}{13}\right) + \cos\left(\frac{5\pi}{13}\right)$$

$$\Rightarrow \text{LHS} = \cos\left(\frac{10\pi}{13}\right) + \cos\left(\frac{8\pi}{13}\right) + \cos\left(\frac{3\pi}{13}\right) + \cos\left(\frac{5\pi}{13}\right)$$

$$\Rightarrow \text{LHS} = \cos\left(\pi - \frac{3\pi}{13}\right) + \cos\left(\pi - \frac{5\pi}{13}\right) + \cos\left(\frac{3\pi}{13}\right) + \cos\left(\frac{5\pi}{13}\right)$$

$$\Rightarrow \text{LHS} = -\cancel{\cos\left(\frac{3\pi}{13}\right)} - \cancel{\cos\left(\frac{5\pi}{13}\right)} + \cancel{\cos\left(\frac{3\pi}{13}\right)} + \cancel{\cos\left(\frac{5\pi}{13}\right)}$$

$$\Rightarrow \text{LHS} = 0 = \text{RHS}$$

Hence proved.

QUESTION

$$* \sin x + \sin y = 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$$

$$* \cos x - \cos y = -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

Prove that:

$$(\sin 3x + \sin x) \sin x + (\cos 3x - \cos x) \cos x = 0$$

Proof:

$$\text{LHS} = \left(2 \sin\left(\frac{3x+x}{2}\right) \cos\left(\frac{3x-x}{2}\right)\right) \sin x + \left(-2 \sin\left(\frac{3x+x}{2}\right) \sin\left(\frac{3x-x}{2}\right)\right) \cos x$$

$$\Rightarrow \text{LHS} = \cancel{2 \sin 2x \cos x \sin x} - \cancel{2 \sin 2x \sin x \cos x}$$

$$\Rightarrow \boxed{\text{LHS} = 0 = \text{RHS}}$$

Hence proved.

QUESTION

$$\begin{aligned} & \cos 2x = 2\cos^2 x - 1 \\ & \cos x = 2\cos^2 \frac{x}{2} - 1 \end{aligned}$$

Prove that:

$$(\cos x + \cos y)^2 + (\sin x - \sin y)^2 = 4\cos^2 \frac{x+y}{2}$$

Proof: LHS = $\cos^2 x + \cos^2 y + 2\cos x \cos y$
 $+ \sin^2 x + \sin^2 y - 2\sin x \sin y$

$$\Rightarrow \text{LHS} = (\cos^2 x + \sin^2 x) + (\cos^2 y + \sin^2 y) + 2(\cos x \cos y - \sin x \sin y)$$

$$\Rightarrow \text{LHS} = 1 + 1 + 2\cos(x+y)$$

$$\Rightarrow \text{LHS} = 2 + 2\cos(x+y)$$

$$\Rightarrow \text{LHS} = 2(1 + \cos(x+y))$$

$$\Rightarrow \text{LHS} = 2\left(1 + 2\cos^2\left(\frac{x+y}{2}\right) - 1\right)$$

$$\Rightarrow \text{LHS} = 4\cos^2\left(\frac{x+y}{2}\right) = \text{RHS.}$$

Hence proved.

QUESTION

$$\cos 2x = 1 - 2\sin^2 x$$

$$\cos x = 1 - 2\sin^2 \frac{x}{2}$$

Prove that:

$$(\cos x - \cos y)^2 + (\sin x - \sin y)^2 = 4\sin^2 \frac{x - y}{2}$$

(Homework Problem)

* Post in on our
* Community

QUESTION



Prove that:

$$\sin x + \sin 3x + \sin 5x + \sin 7x = 4 \cos x \cos 2x \sin 4x$$

Proof: LHS = $(\sin x + \sin 7x) + (\sin 3x + \sin 5x)$

$$\Rightarrow \text{LHS} = 2 \sin\left(\frac{x+7x}{2}\right) \cos\left(\frac{x-7x}{2}\right) + 2 \sin\left(\frac{3x+5x}{2}\right) \cos\left(\frac{3x-5x}{2}\right)$$

$$\Rightarrow \text{LHS} = 2 \sin 4x \cos(-3x) + 2 \sin 4x \cos(-x)$$

$$\Rightarrow \text{LHS} = 2 \sin 4x (\cos 3x + \cos x)$$

$$\Rightarrow \text{LHS} = 2 \sin 4x \times 2 \cos\left(\frac{3x+x}{2}\right) \cos\left(\frac{3x-x}{2}\right)$$

$$\Rightarrow \text{LHS} = 4 \sin 4x \cos 2x \cos x$$

$$\Rightarrow \text{LHS} = 4 \cos x \cos 2x \sin 4x = \text{RHS}$$

Hence proved

QUESTION

$$\sin x + \sin y = 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$$

Prove that:

$$\cos x + \cos y = 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$$

$$\frac{(\sin 7x + \sin 5x) + (\sin 9x + \sin 3x)}{(\cos 7x + \cos 5x) + (\cos 9x + \cos 3x)} = \tan 6x$$

Proof: LHS = $\frac{2 \sin\left(\frac{7x+5x}{2}\right) \cos\left(\frac{7x-5x}{2}\right) + 2 \sin\left(\frac{9x+3x}{2}\right) \cos\left(\frac{9x-3x}{2}\right)}{2 \cos\left(\frac{7x+5x}{2}\right) \cos\left(\frac{7x-5x}{2}\right) + 2 \cos\left(\frac{9x+3x}{2}\right) \cos\left(\frac{9x-3x}{2}\right)}$

$$\Rightarrow \text{LHS} = \frac{\sin 6x \cos x + \sin 6x \cos 3x}{\cos 6x \cos x + \cos 6x \cos 3x}$$

$$\Rightarrow \text{LHS} = \frac{\sin 6x (\cancel{\cos x} + \cos 3x)}{\cos 6x (\cancel{\cos x} + \cos 3x)}$$

$$\Rightarrow \text{LHS} = \tan 6x = \text{RHS}$$

hence proved.

QUESTION

$$* \sin 2x = 2 \sin x \cos x$$

$$* \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$* \sin x + \sin y = 2 \sin \left(\frac{x+y}{2} \right) \cos \left(\frac{x-y}{2} \right)$$

$$* \sin x - \sin y = 2 \cos \left(\frac{x+y}{2} \right) \sin \left(\frac{x-y}{2} \right)$$

Prove that:

$$\sin 3x + \sin 2x - \sin x = 4 \sin x \cos \frac{x}{2} \cos \frac{3x}{2}$$

Proof: LHS = $\sin 3x + (\sin 2x - \sin x)$

$$\Rightarrow \text{LHS} = \sin 3x + 2 \cos \left(\frac{2x+x}{2} \right) \sin \left(\frac{2x-x}{2} \right)$$

$$\Rightarrow \text{LHS} = \sin \left[2 \left(\frac{3x}{2} \right) \right] + 2 \cos \left(\frac{3x}{2} \right) \sin \left(\frac{x}{2} \right)$$

$$\Rightarrow \text{LHS} = 2 \sin \left(\frac{3x}{2} \right) \cos \left(\frac{3x}{2} \right) + 2 \cos \left(\frac{3x}{2} \right) \sin \left(\frac{x}{2} \right)$$

$$\Rightarrow \text{LHS} = 2 \cos \left(\frac{3x}{2} \right) \left[\sin \left(\frac{3x}{2} \right) + \sin \left(\frac{x}{2} \right) \right]$$

$$\Rightarrow \text{LHS} = 2 \cos \left(\frac{3x}{2} \right) \times 2 \sin \left(\frac{\frac{3x}{2} + \frac{x}{2}}{2} \right) \cos \left(\frac{\frac{3x}{2} - \frac{x}{2}}{2} \right)$$

$$\Rightarrow \text{LHS} = 2 \cos \left(\frac{3x}{2} \right) \times 2 \sin \left(\frac{2x}{2} \right) \cos \left(\frac{x}{2} \right)$$

$$\Rightarrow \text{LHS} = 4 \sin x \cos \left(\frac{x}{2} \right) \cos \left(\frac{3x}{2} \right) = \text{RHS}$$

Hence proved.

QUESTION

(Important) (P/Q) * $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

* $\tan x = \frac{2 \tan(x/2)}{1 - \tan^2(x/2)}$

Find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$ in each of the following:

$\tan x = -\frac{4}{3}$, x in quadrant II

(Sol) we know that $\tan x = \frac{2 \tan(x/2)}{1 - \tan^2(x/2)}$

$\Rightarrow -\frac{4}{3} = \frac{2 \tan(x/2)}{1 - \tan^2(x/2)}$

$\Rightarrow -\frac{4}{3} = \frac{2t}{1-t^2}$, where $t = \tan(x/2)$

$\Rightarrow -4(1-t^2) = 6t$

$\Rightarrow -2 + 2t^2 = 3t$

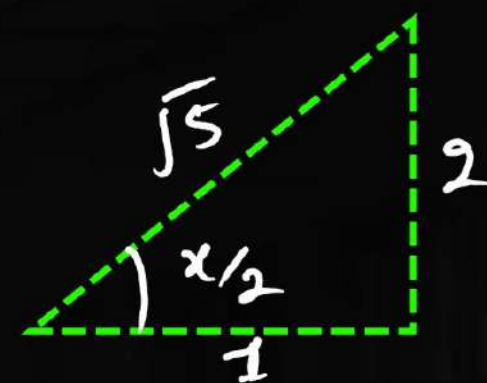
$\Rightarrow 2t^2 - 3t - 2 = 0$

$\Rightarrow 2t^2 - 4t + t - 2 = 0$

$\Rightarrow 2t(t-2) + 1(t-2) = 0$ So, $\tan(x/2) = 2$

$\Rightarrow (2t+1)(t-2) = 0$

$\Rightarrow t = -\frac{1}{2}, 2$



$\Rightarrow \tan(x/2) = -\frac{1}{2}, 2$

As $\frac{\pi}{2} < x < \pi$

$\Rightarrow \frac{\pi}{4} < \frac{x}{2} < \frac{\pi}{2}$

$\therefore \sin(x/2) = \frac{2}{\sqrt{5}}$
 $\cos(x/2) = \frac{1}{\sqrt{5}}$

QUESTION

$$\cos 2x = 2\cos^2 x - 1$$
$$\cos x = 2\cos^2\left(\frac{x}{2}\right) - 1.$$

Find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$ in each of the following:

$$\cos x = -\frac{1}{3}, x \text{ in quadrant III}$$

(Sol) we know that $\cos x = 2\cos^2\left(\frac{x}{2}\right) - 1$.

$$\Rightarrow -\frac{1}{3} + 1 = 2\cos^2\left(\frac{x}{2}\right)$$

$$\Rightarrow \cancel{2}\cos^2\left(\frac{x}{2}\right) = \cancel{2}\frac{2}{3}$$

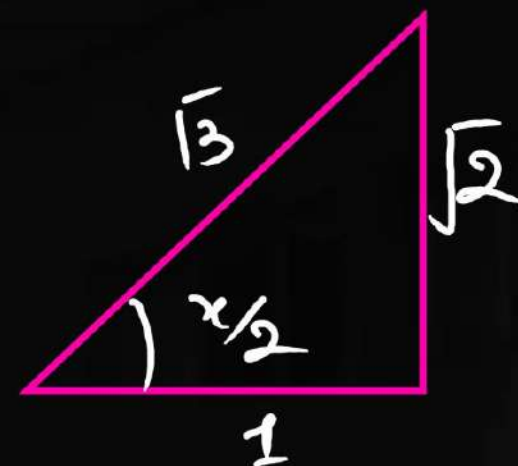
$$\Rightarrow \cos^2\left(\frac{x}{2}\right) = \frac{1}{3}.$$

$$\Rightarrow \cos\left(\frac{x}{2}\right) = \pm \frac{1}{\sqrt{3}}.$$

$$\text{As } \pi < x < \frac{3\pi}{2}$$

$$\Rightarrow \frac{\pi}{2} < \frac{x}{2} < \frac{3\pi}{4}$$

$$\text{So, } \cos\left(\frac{x}{2}\right) = -\frac{1}{\sqrt{3}}$$



$$\text{Therefore } \sin\left(\frac{x}{2}\right) = \frac{\sqrt{2}}{\sqrt{3}}.$$

$$\tan\left(\frac{x}{2}\right) = -\sqrt{2}.$$

Hence the result.

QUESTION

Find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$ in each of the following:

$$\sin x = \frac{1}{4}, x \text{ in quadrant II}$$

$$\underline{\cos^2 x + \sin^2 x = 1} \quad (\text{Homework Problem})$$

$$\cos x = 2\cos^2\left(\frac{x}{2}\right) - 1$$

QUESTION

Maximum value of $\sin x + \cos x$ is

$$a=1, b=1$$

$$\text{Max. value} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

☒ $\sqrt{2}$

☐ 1

☐ 2

☐ $\frac{1}{\sqrt{2}}$

$$E = a \sin x + b \cos x.$$

$$E_{\max} = \sqrt{a^2 + b^2}$$

$$E_{\min} = -\sqrt{a^2 + b^2}$$

QUESTION

$$\frac{9}{2} + \frac{9}{2} = 2 \times \frac{9}{2}$$



The Maximum value of $1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2\cos\left(\frac{\pi}{4} - \theta\right)$ is

☒ 1

☒ 2

☒ 3

☒ 4

(Sol) let $y = 1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2\cos\left(\frac{\pi}{4} - \theta\right)$

$$\Rightarrow y = 1 + \sin\frac{\pi}{4}\cos\theta + \cos\frac{\pi}{4}\sin\theta + 2\left(\cos\frac{\pi}{4}\cos\theta + \sin\frac{\pi}{4}\sin\theta\right)$$

$$\Rightarrow y = 1 + \frac{\cos\theta}{\sqrt{2}} + \frac{\sin\theta}{\sqrt{2}} + \frac{2}{\sqrt{2}}\cos\theta + \frac{2\sin\theta}{\sqrt{2}}$$

$$\Rightarrow y = 1 + \left(\frac{3}{\sqrt{2}}\sin\theta + \frac{3}{\sqrt{2}}\cos\theta\right)$$

$$\Rightarrow y_{\max} = 1 + \left(\frac{3}{\sqrt{2}}\sin\theta + \frac{3}{\sqrt{2}}\cos\theta\right)_{\max}$$

$$\Rightarrow y_{\max} = 1 + \sqrt{\frac{3^2}{\sqrt{2}^2} + \frac{3^2}{\sqrt{2}^2}}$$

$$\Rightarrow y_{\max} = 1 + \sqrt{\frac{9}{2} + \frac{9}{2}}$$

$$\Rightarrow y_{\max} = 1 + \sqrt{9} = 1 + 3$$

$$\Rightarrow y_{\max} = 4$$

QUESTION

The Statement $\cos \theta = \frac{1+t^2}{1-t^2}$ is valid for

1 All real t

2 All $t > 0$

3 $t = 0$

4 None of these

Homework.

Thank
You